Pricing Optimization for Burger Gram

A Final Submission report for the BDM capstone Project

Submitted by

Name: Jeet Kumar

Roll number: 21f3002041

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Declaration Statement

I am working on a Project titled “Pricing Optimization for Burger Gram”. I extend my appreciation to **Burger Gram**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate

Name: Jeet Kumar

Date: 15-Feb-2024

Contents

1 Executive Summary and Title 3

2 Detailed Explanation of Analysis Process/ Method 4

3 Results and Findings 8

Sales Analysis and Revenue Analysis of Burger Gram: 8

4 Interpretation of Results and Recommendation 15

1 Executive Summary and Title

Burger Gram, a small-scale fast-food shop in Sarhaul Village, Sector-18, Gurugram, Haryana, operates as a B2C business with a focus on delivering high-quality, affordable food. Despite its commitment to quality, the business faces challenges related to profit management and inventory control, hindering growth.

The capstone project aims to analyze menu item performance and implement changes to reduce raw material wastage and enhance planning. By optimizing product prices, the project seeks to boost overall profitability.

Key Challenges:

1. Raw Material Costs: Efficiently managing storage and handling costs for raw materials and finished products.
2. Low Demand: Addressing products with insufficient sales and revenue.

To tackle these issues, I will analyze one month of sales data to identify the best and worst-performing food categories and items. Insights from this analysis will guide business strategies aimed at improving net sales and increasing profits.

Data Analytics Approach:

- Leveraging Excel tools (graphs, charts, pivot tables) to visualize patterns and sales trends.
- Adjusting product pricing to optimize revenue and expenses.
- Removing underperforming products to reduce inventory costs.

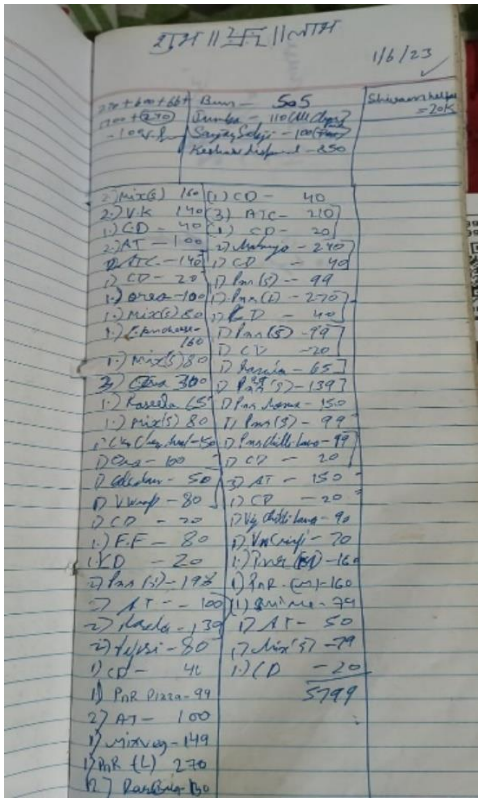
However, it’s essential to acknowledge the project’s limitations:

1. Sample Size: The analysis relies on a single month’s data; a broader dataset would yield more robust insights.
2. Assumptions: Sales patterns may not remain constant due to external factors.
3. Market Dynamics: Competitive analysis and market trends are not considered, impacting decision-making.

2 Detailed Explanation of Analysis Process/ Method

A. Data analysis for sales

I've gathered sales data for 30 days, spanning from June 1, 2023, to June 30, 2023. The analysis will be conducted using Excel, where the data will be entered into a spreadsheet. Initially, this raw data will undergo cleaning and reorganization to ensure its accuracy and usability.



Informal way of storing sales data

The prepared data has menu sheet and the sales data sheet –

S. No.	Food Item	Category	Price (in Rupees)
1	Aloo Tikki Burger	Burger	50.00
2	Aloo Tikki Supreme Burger	Burger	60.00
3	Veg Burger	Burger	60.00
4	Aloo Tikki Cheese Burger	Burger	70.00
5	Veg King Burger (with Cheese)	Burger	70.00
6	Veg 'N' Crisp Burger	Burger	70.00
7	Veg Surprise Burger	Burger	90.00
8	Veg Chilli Lava Burger	Burger	90.00
9	Veg Thick Burger (with Cheese)	Burger	90.00
10	Double Veg 'N' Crisp Burger	Burger	99.00
11	Veg Crisp 'N' Crunchy Burger	Burger	99.00
12	Maharaja Supreme Burger	Burger	120.00
13	Paneer Tikki Burger	Burger	90.00
14	Paneer Chilli Lava Burger	Burger	99.00
15	Paneer Crisp 'N' Crunchy Burger	Burger	99.00

1- Burger Gram's menu made available in spreadsheet format

SN	Quanti	Food Product	Category	Pri	Total Pai	Date
1	2	Mix Veg Pizza (Small)	Pizza	79	158	01 June 2023
2	2	Veg King Burger (with Cheese)	Burger	70	140	01 June 2023
3	1	Cold Drink	Cold Drink	40	40	01 June 2023
4	2	Aloo Tikki Burger	Burger	50	100	01 June 2023
5	2	Aloo Tikki Cheese Burger	Burger	70	140	01 June 2023
6	1	Cold Drink (Small)	Cold Drink	20	20	01 June 2023
7	1	Oreo Shake	Shake	100	100	01 June 2023
8	1	Mix Veg Pizza (Small)	Pizza	79	79	01 June 2023
9	1	Cheese Corn Nuggets (12pcs)	Fries	160	160	01 June 2023
10	1	Mix Veg Pizza (Small)	Pizza	79	79	01 June 2023
11	3	Oreo Shake	Shake	100	300	01 June 2023
12	1	Veg Raseela Sandwich	Sandwich	65	65	01 June 2023
13	1	Mix Veg Pizza (Small)	Pizza	79	79	01 June 2023
14	1	Chicken Kurkure Momos	Momos	150	150	01 June 2023
15	1	Oreo Shake	Shake	100	100	01 June 2023
16	1	Veg Coleslaw S/W	Sandwich	50	50	01 June 2023

2- Burger Gram's sales data prepared in Excel

The preprocessed data has 7 columns as shown above. The Revenue can be easily found with the formula:

Revenue = Selling Price * Sales

Total Revenue = Summation of all Revenues for each product.

B. Profit increase:

As told by the owner he was ready to make modifications in the prices of the food items based on their demand and popularity.

So first we will find the total 30-day sales for every food item and check for those who are performing the best and those who are not performing well by the help of pivot tables.

Food Items	Sales
Aloo Tikki Burger	340
Mix Veg Pizza (Small)	267
Cold Drink (Small)	195
Paneer Pizza (Small)	115
Cold Coffee Shake	106
Veg Raseela Sandwich	105
Aloo Tikki Supreme Burger	81
Paneer Pizza (Medium)	80
Mix Veg Pizza (Medium)	60
Sweet Corn Pizza (Small)	58
Veg Burger	53
Cold Drink	53
Aloo Tikki Cheese Burger	50
Paneer Thick Wrap	49
Extra Cheese	34

\

Then, we will find the total revenue created by each food item.

Product	Revenue by Product
Mix Veg Pizza (Small)	12.43%
Aloo Tikki Burger	10.02%
Paneer Pizza (Medium)	7.54%
Paneer Pizza (Small)	6.71%
Mix Veg Pizza (Medium)	5.27%
Cold Coffee Shake	4.37%
Veg Raseela Sandwich	4.02%
Aloo Tikki Supreme Burger	2.86%
Sweet Corn Pizza (Small)	2.73%
Paneer Thick Wrap	2.60%
Cold Drink (Small)	2.30%
Aloo Tikki Cheese Burger	2.06%
Veg Burger	1.87%

This sorting of data on the basis of their contribution in sales and revenue will help the owner to make data informed decisions. So those which are performing the best and are in high demand can have small price increase creating more profit to the

business.

C. Optimizing the inventory:

The owner grapples with significant raw material wastage due to underperforming menu items, making efficient raw material management a challenge. As per the owner's instructions, my task is to identify items with minimal or no demand. By analyzing sales data, we can readily pinpoint products with zero or very low sales, allowing us to streamline the menu and minimize wastage.

The categories which perform very poor can be found by checking minimum contributions in sales after applying filter feature of excel. We can clearly see that these are the food categories which have sales less than 5% contribution in total sales, so are poor performing categories.

Wrap	4.23%	4.68%
Non Veg Burger	2.99%	3.24%
Momos	2.44%	4.35%
Extra Cheese	1.75%	0.97%
Choco Lava	0.09%	0.09%

Also, it can be checked product wise means which food product has minimum sales by sorting it in descending or ascending order.

Least Selling	Sweet Corn Pizza (Large)	0.05%
	Veg Crisp 'N' Crunchy Burger	0.05%
	Chicken Surprise Burger	0.05%
	Chicken Crisp 'N' Crunchy Burger	0.05%
	Egg Burger	0.05%
	Loaded French Fries	0.05%
	Chicken 'N' Crisp Burger	0.05%

And the most important, finding those products which have zero contribution in shop's growth, just by applying VLOOKUP formula in the menu sheet and selecting the table array from sales sheet. This will just show '#N/A' in front of those items which are in menu but not in sales data.

S. No.	Food Item	Category	Price (in Rupees)	Sales
5	Veg King Burger (with Cheese)	Burger	70.00	#N/A
6	Veg 'N' Crisp Burger	Burger	70.00	60
7	Veg Surprise Burger	Burger	90.00	280
8	Veg Chilli Lava Burger	Burger	90.00	80
9	Veg Thick Burger (with Cheese)	Burger	90.00	80
10	Double Veg 'N' Crisp Burger	Burger	99.00	100

By looking at Sales column, it can be observed that Veg King Burger (with Cheese) has no sales value so indicating #NA.

3 Results and Findings

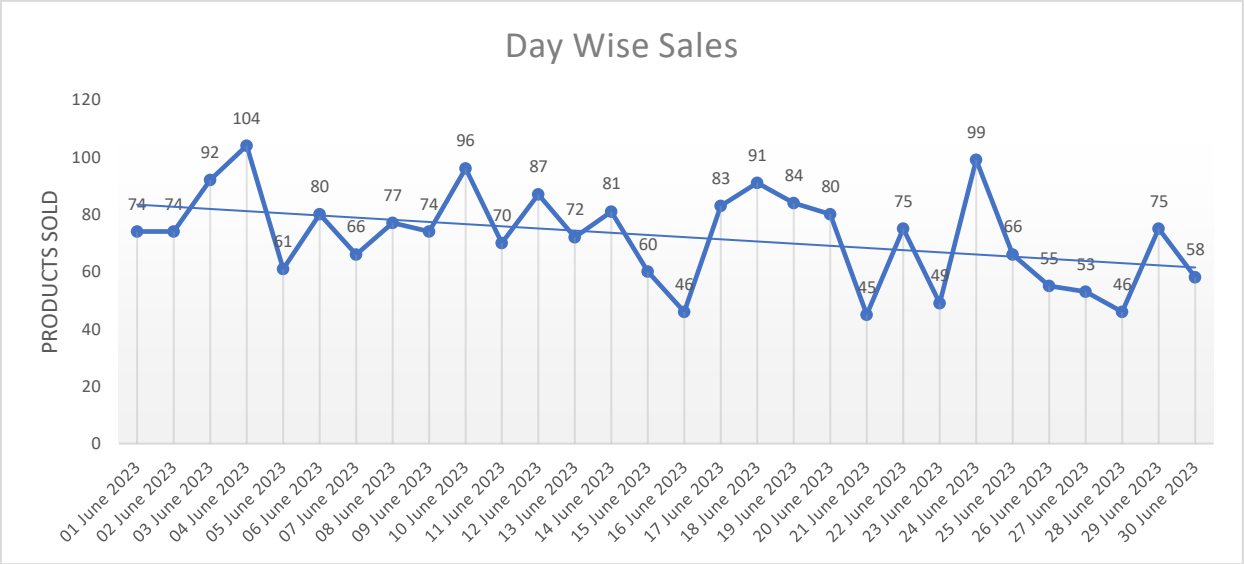
Sales Analysis and Revenue Analysis of Burger Gram:

Day wise sales of various food items-

Line chart is used here to show day wise sales data. Here one can easily see the total sales for a day and 4/June/23 is the day of the month when the business recorded the highest sales i.e. 104. Also, the minimum goes to 46 for 28/June/23. The trendline shows the overall decrease in sales.

This analysis shows that average daily sales is 72.43

Also, the grand total for the June/23 is 2173



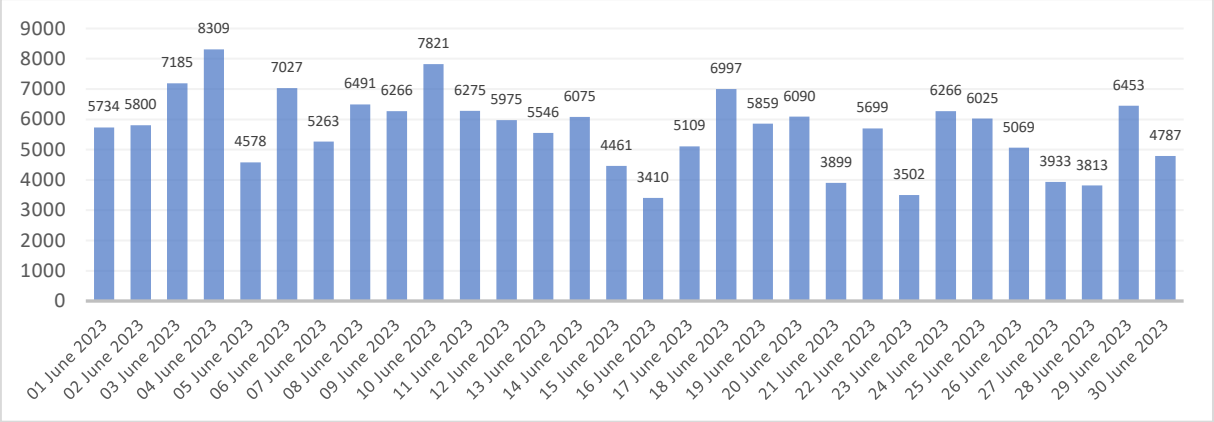
Day wise revenue-

here in this column chart is show to indicate the revenue per day to the business. 4/June/23 marks the highest revenue of Rs. 8309 and the lowest goes to 16/June/23 with only Rs. 3410.

Average Daily revenue is Rs. 5657.23

And Grand Total of revenue is Rs. 169717

Range = Max-Min = 8309- 3410 = Rs. 4899

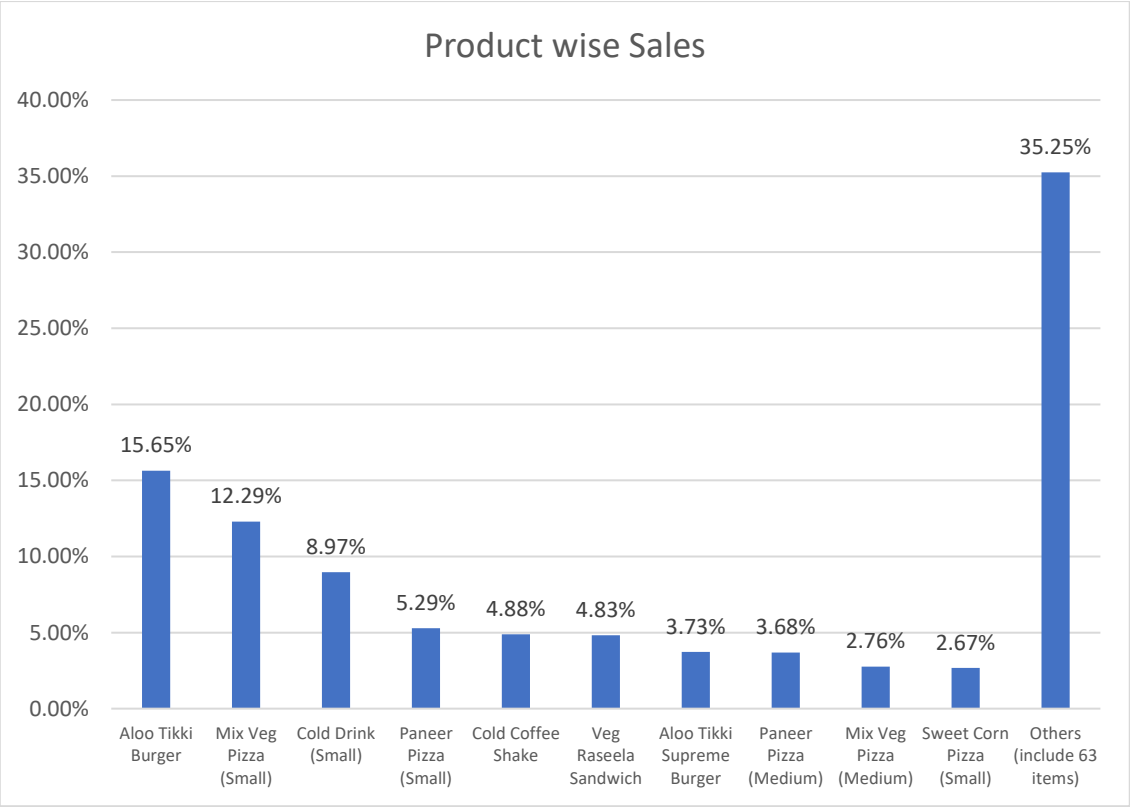


It can be seen from above two graphs, the revenue and sales are directly related meaning when sales increases, revenue also increases.

Product wise sales-

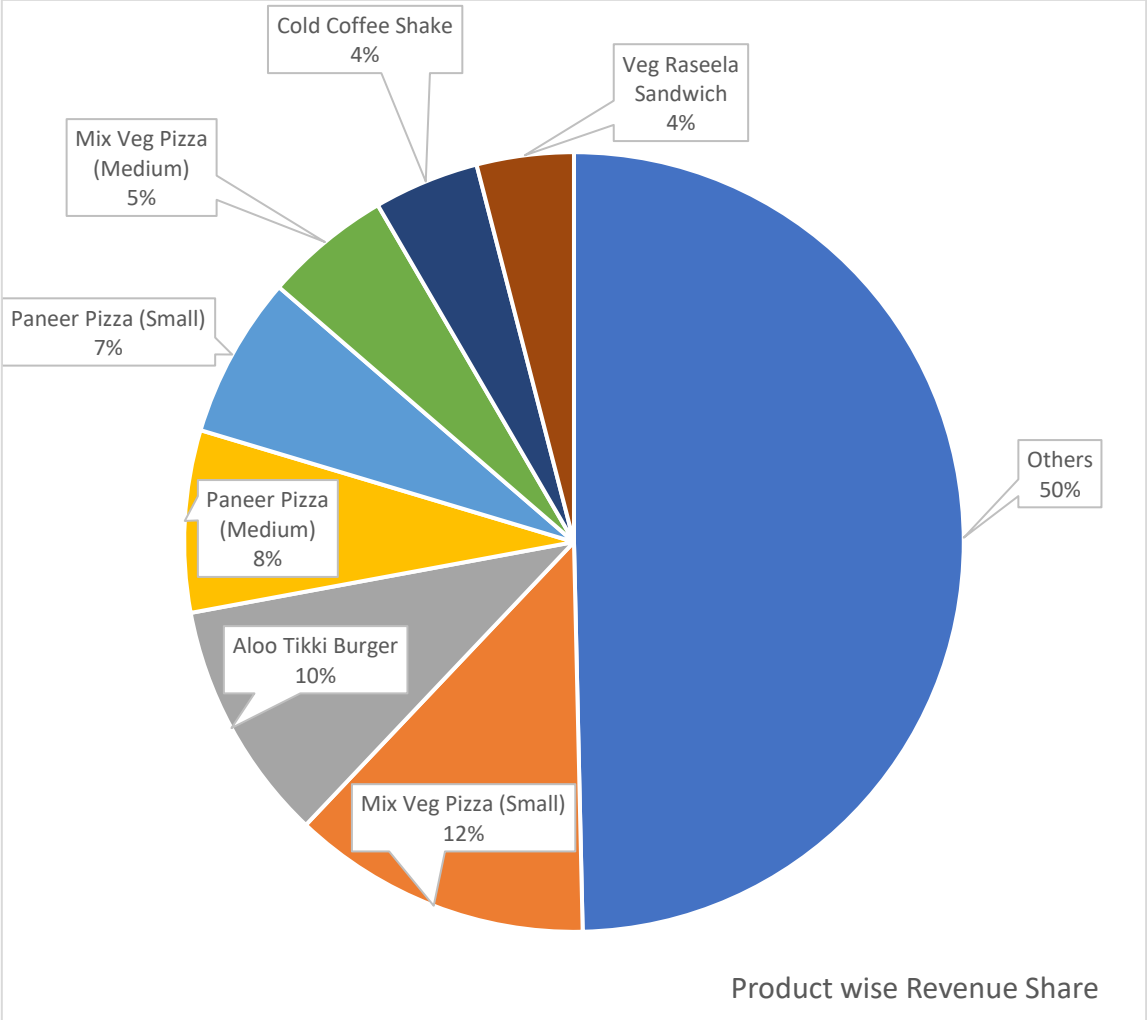
The product wise sales indicate the sales data for each food product over a period of 30 days. Aloo Tikki and Mix Veg Pizza are two most selling products. Here Others include 63 items which are having comparatively lower sales (below 2.5%).

The Aloo Tikki Burger which costs Rs. 50 and Mix Veg Pizza (Small) which costs Rs. 79 are in high demand when combined have 28% sales with just two food items. It also shows that big section of customers prefer low price products in specific category such as Burgers and Pizzas.



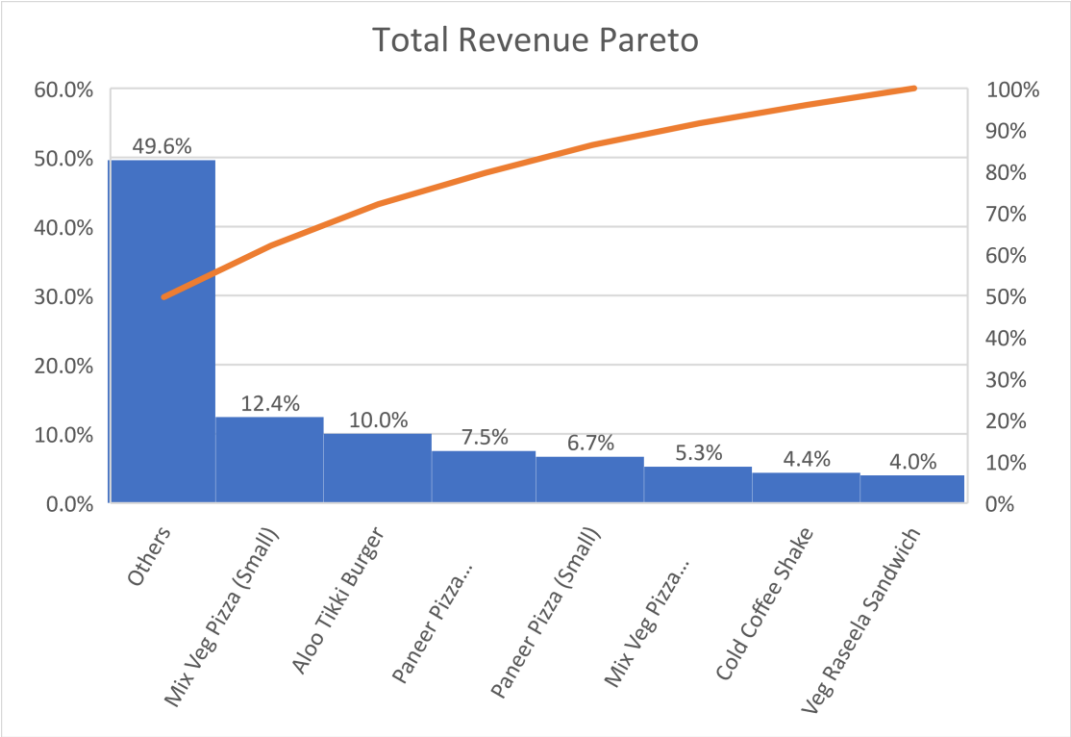
Product wise revenue-

The visualization below is a pie chart. Each slice corresponds to food items' contribution. Others include 66 items which are having comparatively lower sales (below 4%).



Pareto Chart for Total Revenue-

Pareto chart can easily show that the best-selling products are the Mix Veg Pizza (Small), Aloo Tikki Burger, Paneer Pizza (Medium), Paneer Pizza (Small), Mix Veg Pizza (Medium), Cold coffee shake, Veg Raseela Sandwich. Only these 7 contribute in huge amount to the total revenue.

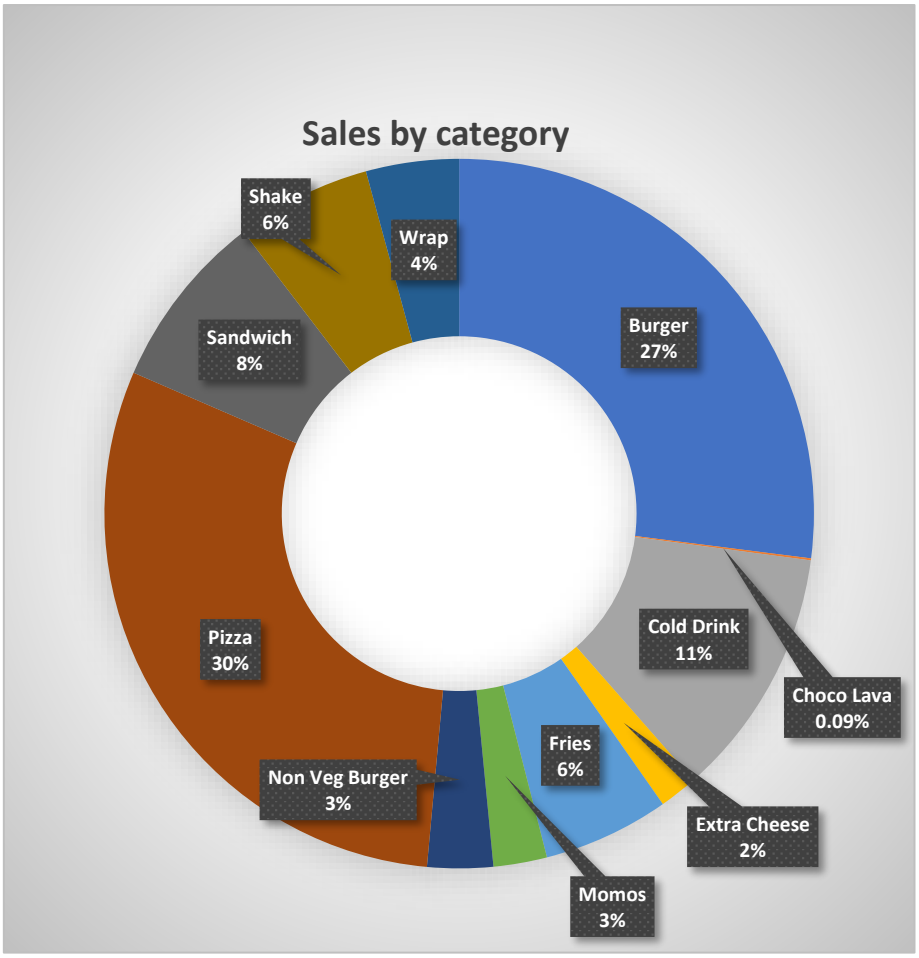


Category wise sales-

The Donut pie chart below indicates that Pizza is the most preferred category followed by Burgers, then many customers prefer Cold Drinks, it can be said that cold drinks are also bought by the customers when they buy Burgers or Pizzas.

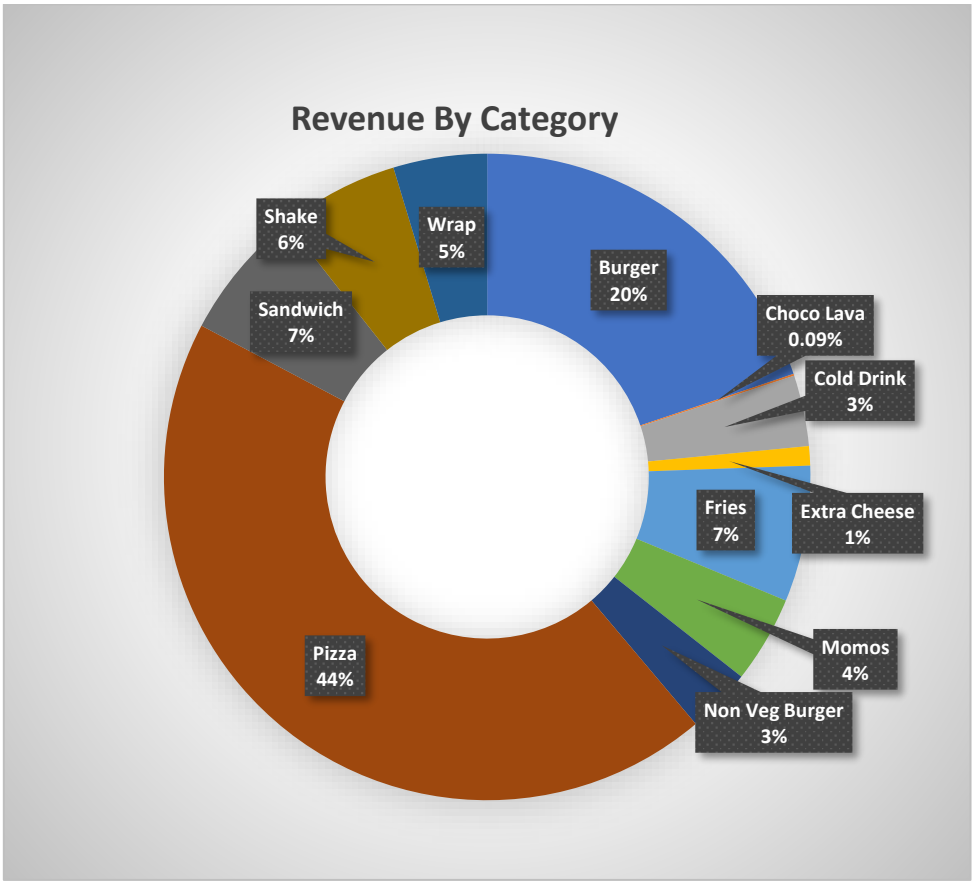
Choco Lava sales are very low i.e. 0.09%. This indicates that customers rarely opt for Choco lava.

It can be noted that Aloo Tikki burger is the most selling product, then also Pizza is the most famous as a category that means, there are other pizzas which are performing well but Aloo Tikki Burger is the only performing player in its category.



Category wise revenue-

This donut pie chart shows the revenue contributions from each category to the business. The Pizza category alone has 44% of total revenue contributions, this indicates that pizzas are the high performing players and they bring large revenue from low sales volume than Burgers which have only 20% contribution when have 27% sales volume.



4 Interpretation of Results and Recommendation

Recommendations:

1. Increasing the sales of high profit item:
   As we have seen in the analysis, Pizzas are very attractive to customers as well as bring most of the revenue, hence high net profit to the business.

   Steps which can be taken to increase sales:
   B2B partnerships: Pizzas can be easily sold to businesses which organize events and parties on special occasions. Burger Gram can tie up with the event organizers and ensure sales of pizzas in bulk.

   Promotions: Burger Gram can opt for promotions for their products which will result in attracting more customers and business can grow much more.

   Discounts: The discounts are a good way to offer food items to customers. This way they can increase the number of customers also, the customers can repeat.

2. Increasing the price of product in high demand:
   We have seen that the Burgers especially, Aloo Tikki Burger alone contributes in sales by approximately 16%. Also, The Mix Veg Pizza Small size and cold drinks have almost 21% share when combined. These items are very cheap in price presently but in high demand. Just by introducing a nominal increase in the price of these three items will increase the net profit.

3. Removing the unnecessary load:
   There are many products which are not working efficiently and have very low or poor performance but require equal attention.

Sweet Corn Pizza (Large)	0.05%
Veg Crisp 'N' Crunchy Burger	0.05%
Chicken Surprise Burger	0.05%
Chicken Crisp 'N' Crunchy Burger	0.05%
Egg Burger	0.05%
Loaded French Fries	0.05%
Chicken 'N' Crisp Burger	0.05%

The worst-performing items currently account for a mere 0.35% of the total sales. The owner's objective is to streamline the menu, creating a concise selection that allows for focused attention and increased profitability. By eliminating these underperforming items, the business can achieve two significant benefits:

Minimal Loss: Since these items contribute insignificantly to overall sales, their removal won't result in substantial revenue loss. Instead, it will free up resources and reduce operational costs.

Raw Material Savings: By discontinuing these low-demand products, the shop can curb raw material wastage. This prudent move ensures efficient inventory management and cost savings.

In summary, trimming the menu strategically will enhance efficiency, boost profits, and minimize unnecessary burdens.

Also, eliminating those which have zero sales:

Based on our data analysis, we've identified seven items that contribute negligibly to business growth. These items exhibit zero sales and have failed to resonate with any customers. Continuing to offer them provides no advantage to the business.

Here's why removing these items makes sense:

Business Efficiency: By discontinuing these non-performing products, the business can streamline operations. This frees up resources and allows the team to focus on more profitable ventures.

Customer Satisfaction: Eliminating unpopular items ensures that customers encounter a curated menu with offerings they appreciate. Happy customers are more likely to return and recommend the shop to others.

Cost Savings: These items contribute nothing to revenue but still incur raw material costs. Removing them reduces wastage and improves overall cost management.

In summary, letting go of these unproductive menu items is a strategic move that aligns with business growth and customer satisfaction.

Food Item	Category	Price (in Rupees)	Sold
Veg Thick Burger (with Cheese)	Burger	90	0
Double Veg 'N' Crisp Burger	Burger	99	0
Veg Maxican Sandwich	Sandwich	60	0
Burger Gram Spl. Sandwich Veg	Sandwich	100	0
Burger Gram Spl. Sandwich Non Veg	Sandwich	120	0
Chicken Thick Burger	Non Veg Burger	100	0
Fresh Lime	Shake	60	0
Chicken Pizza (Large)	Pizza	349	0

4. Effective Pricing:
   Combo offers like Pizza+ French Fries+ Cold Drink can be introduced at a discounted price so that customers buy multiple items and there will be overall increase in sales.

   Promotions: Promotions in fast food market like buy-one get-one, limited time offers and discounts etc. are very popular and they actually work. This will ensure growth of the business.

5. Keeping track of data:
   Beyond sales records, the owner should track all financial aspects of the business. This includes costs, inventory, salaries, rent, and other expenses.
   Maintaining a spreadsheet format for data storage ensures easy access and historical analysis. It allows the owner to identify trends, cost fluctuations, and areas for improvement.

Customer Feedback Loop:

Introducing a voluntary feedback form for customers is a smart move. It provides valuable insights into their experiences, preferences, and any areas needing improvement.

Regularly reviewing this feedback allows the business to address customer perspectives promptly. Whether it's enhancing service quality, adjusting menu items, or improving ambiance, direct feedback guides decision-making.

Customer Database:

Collecting and organizing customer details (such as names and phone numbers) creates a loyalty database.

Benefits:

Repeat Business: Recognizing returning customers fosters loyalty. The business can offer personalized promotions or discounts.

Targeted Marketing: Tailoring marketing efforts based on customer preferences becomes feasible.

Reward Programs: Tracking customer visits allows for rewarding loyalty, encouraging repeat visits.

In summary, these practices enhance business efficiency, customer satisfaction, and long-term growth.